

HYDROELECTRIC PROJECT EL EDÉN

Project proposed by

Central Hydroeléctrica El Edén S.A.S. E.S.P.

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1 PROJECT DETAILS

1.1 Summary Description of the Project

The Hydroelectric Project El Edén (hereafter, the project activity or project) is implemented by Central Hidroeléctrica El Edén S.A.S E.S.P (hereafter, the project proponent).

The project activity is a hydroelectric power plant that utilizes water from the Río La Miel ("La Miel river"), located in the river basin "La Miel" in the east of the Department of Caldas, Colombia (hereafter, the Host Country), about 125 kilometres from the City of Cali close to the municipalities of Pensilvania, Marquetalia and Manzanares.

Before the implementation of the project activity, no power plant is installed at the project site. The energy that will be generated by the project is currently dispatched by other power plants that are connected to the national grid, and that include fossil fuel power plants. Hence, the baseline scenario and the scenario prior to project implementation are the same and the project will help to reduce thermal power generation in the National Interconnected System (SIN¹) by providing renewable energy to the grid.

The project activity is a run-of-river hydroelectric plant that includes the construction and installation of a concrete weir and intake, a desander, channel, forebay/settling tank, high-pressure penstock, powerhouse and discharge channel. Two Pelton turbines with 10.115 MW and two synchronous generators with 11,174 kVA each² will be installed. Each generator will be connected to a transformer to increase the voltage from 6.6 kV to 35 kV at the project site. The installed capacity of the plant at the interconnection point will be around 19.5 MW (this is based on the design and can vary slightly at real conditions).

The electricity generated by the project is expected to be around 104.5 GWh per year, which will be delivered to the SIN. It will help to reduce thermal power generation in the SIN by providing renewable energy to the grid. Thus, it will contribute to sustainability by increasing the share of renewable energy and reducing GHG emissions. The expected average annual emission reductions are 32,102 tCO₂ during a renewable 10-years crediting period.

Moreover, the project activity will facilitate the achievement of electricity demand required for the social and economic development of the country based on a renewable energy source. The project will contribute to the region's sustainable development in the following ways:

- The project will decrease emissions of sulphur oxides (SO_x), nitrogen oxides (NO_x), carbon monoxide, particulate matter, other pollutants, as well as carbon dioxide associated with the combustion of fossil fuels.
- The project will reduce the dependence on fossil fuels, which are non-renewable and limited in supply.
- During construction, the project will provide employment opportunities. To the extent practical, these jobs will be filled by local residents. Besides, employment will also be generated for operation and maintenance activities.

1.2 Sectoral Scope and Project Type

Scope 1 - Energy (renewable/non-renewable)

¹ In Spanish: *Sistema Interconectado Nacional*

² With the capacity factor $\cos(\varphi) = 0.9$, the capacity is 10.0566 MW.

The project consists of only one hydroelectric power plant and is not a grouped project.

1.3 Project Proponent

The project proponent is Central Hidroeléctrica El Edén S.A.S E.S.P, which will be responsible for the construction, operation and maintenance of the project.

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1.4 Other Entities Involved in the Project

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1.5 Project Start Date

The project is expected to be operational on 01/01/2015, which will be the project start date. This will be adjusted in accordance with real operational start or commissioning date.

1.6 Project Crediting Period

The first crediting period is 10 years, from 01/09/2015 to 31/08/2025.

1.7 Project Scale and Estimated GHG Emission Reductions or Removals

Project	x
Large project	

Years	Estimated GHG emission reductions or removals (tCO ₂ e)
2015	32,102
2016	32,102
2017	32,102
2018	32,102
2019	32,102
2020	32,102
2021	32,102
2022	32,102
2023	32,102
2024	32,102
Total estimated ERs	321,024
Total number of crediting years	10
Average annual ERs	32,102

1.8 Description of the Project Activity

The project activity is a run-of-river hydroelectric project with an installed turbine capacity of 20.230 MW (two Pelton turbines with 10.115 MW each).

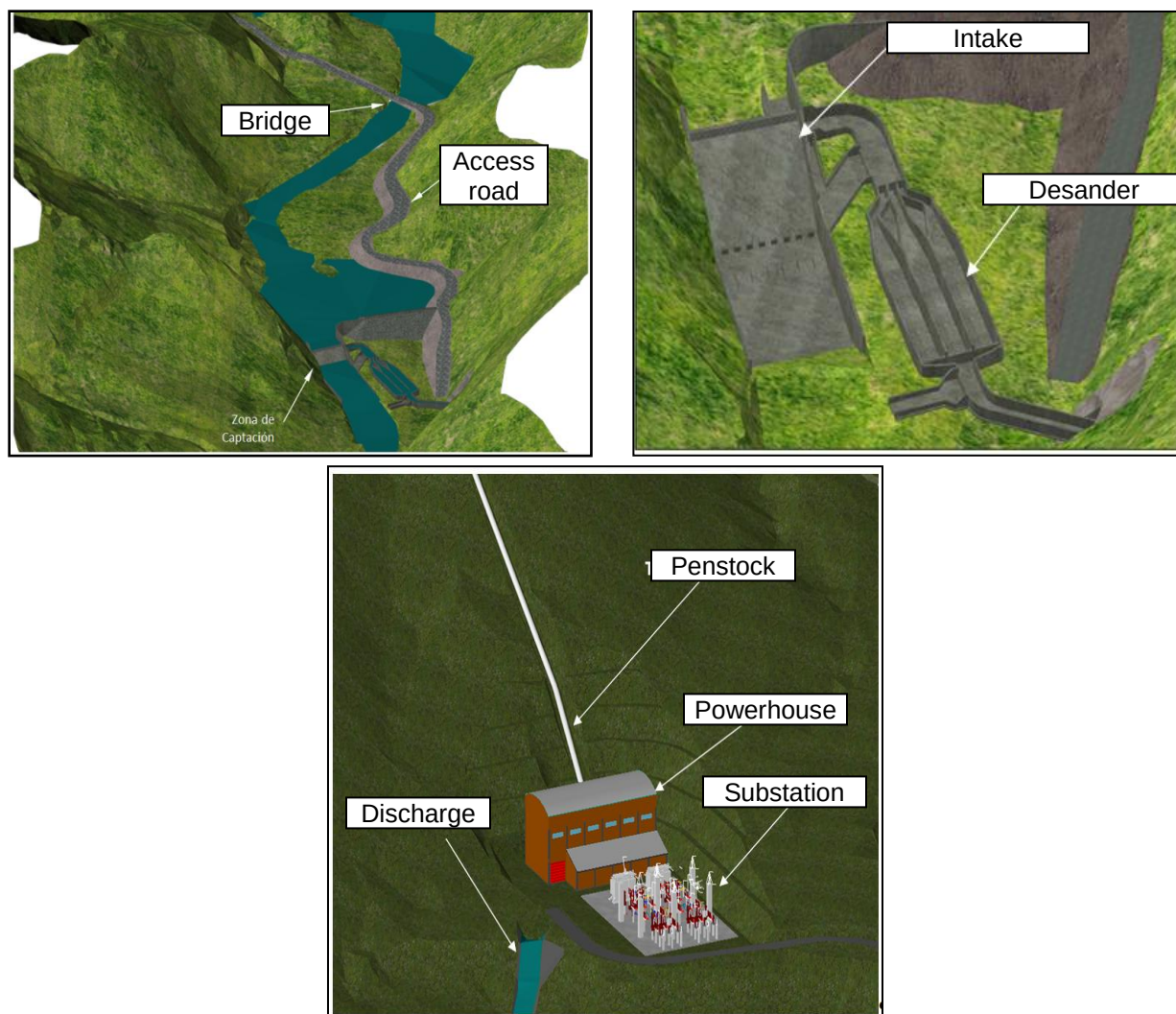


Figure 1: Project scheme showing the main elements of the hydroelectric power plant: Overview of intake area and infrastructure (top left); intake and desander (top right); penstock, powerhouse and discharge (bottom). The project will be connected to an existing substation in the area (it is included in the scheme for completeness, although the substation will not be built as part of the project activity nor will it be located at the same site).

The project includes the following principal infrastructure and equipment:

Weir and intake

A concrete weir and intake structure is used to divert a portion of the river's flow by a river-ground intake. It is composed of concrete weir that creates a small headpond and diversion channel that allows the facility to direct water towards the penstock. The concrete intake is 15 m wide, 14 m long and 5 m high.

Desander

A desander will be constructed in order to retain particles with a size of more than 0.2 mm that could damage and wasting of the turbines. The dimensions of 49 m longitude, 12 m width and 10 m heights are defined based on this size and a design flow of 6.44 m³/s.

Channel and Forebay Tank

From the intake there are three closed channels stages: (i) The first one has a diameter of 2 m and longitude of 908 m; (ii) the second has a diameter of 3.5 m and longitude of 2,115 m; (ii) and the third has a diameter of 2 m and longitude of 1,222 m.

The Forebay tank (settling tank) is build in concrete at the end of the third channel stage before the high-pressure penstock. The tank guarantees that no air enters the penstock and to compensate small variations in short-term water inflow. The dimensions of the tank are 6 m x 10 m, with a height of 7 m.

Penstock

From the Forebay tank, the high-pressure penstock discharges the water to the turbines located in the powerhouse. The penstock consists of two subsequent pipes: (i) the first has a diameter of 1.5 m and a longitude of 639 m (at a pressure of 32 bars) and (ii) the second has a diameter of 1.1 m and a longitude of 65 m (at a pressure greater than 32 bars) .

Powerhouse and discharge channel

The powerhouse is constructed above ground on the left side of the River La Miel. The dimensions are 50 x 10 m in order to guarantee a minimum distance of 11 m between the generation units.

The discharge channel is build with concrete. It will have a width of around 3 m, longitude of 40 m and a height of 1.9 m in order to guarantee that the water can be discharged back to the river.

Turbines and generators

The main equipment consists of two Pelton turbines and two synchronous generators. Additionally, complementary equipment and control devices will be installed.

Table 1: Turbine characteristics (values given per turbine unit).

Description	Value
Number of units	2
Capacity per unit	10,115 kW
Velocity	720 rpm
Frequency	60 Hz
Injection (number)	3
Injection (type)	Internal

Table 2 shows the characteristics of the generators of the project.

Table 2: Generator characteristics (values given per generator unit).

Description	Value
-------------	-------

Number of units	2
Capacity per unit	11,174 kVA
Capacity factor	0.9
Velocity	720 rpm
Voltage	13,800 V
Frequency	60 Hz

Each generator will be connected to a transformer to increase the voltage from 6.6 kV to 35 kV. The transformers will have three phases and work at the same frequency of 60 Hz.

It is important to take into account that the values both tables above (turbines and generators) refer to the current data available from the technical proposal. Typically, the real conditions and possible minor adjustments of the project design during construction can slightly affect the final design at full implementation. However, this does not affect additionality in case that the future values will be in the same range, since (i) the values at the moment of decision making are relevant and shall be taken for the financial analysis; and (ii) such future deviations are expected to be marginal and without a significant impact. In case relevant deviations occur, the project proponent will adjust the data accordingly, at the moment of first verification.

1.9 Project Location

The project is located in the Department of Caldas, Colombia. The project area includes the municipalities of Pensilvania, Marquetalia and Manzanares³.

The coordinates of the intake are E 885038 N 1079832 (datum Bogotá).

³ Environmental Licence, Resolution No. 173 from 04/04/2011, see “Licencia Ambiental El Edén.pdf”

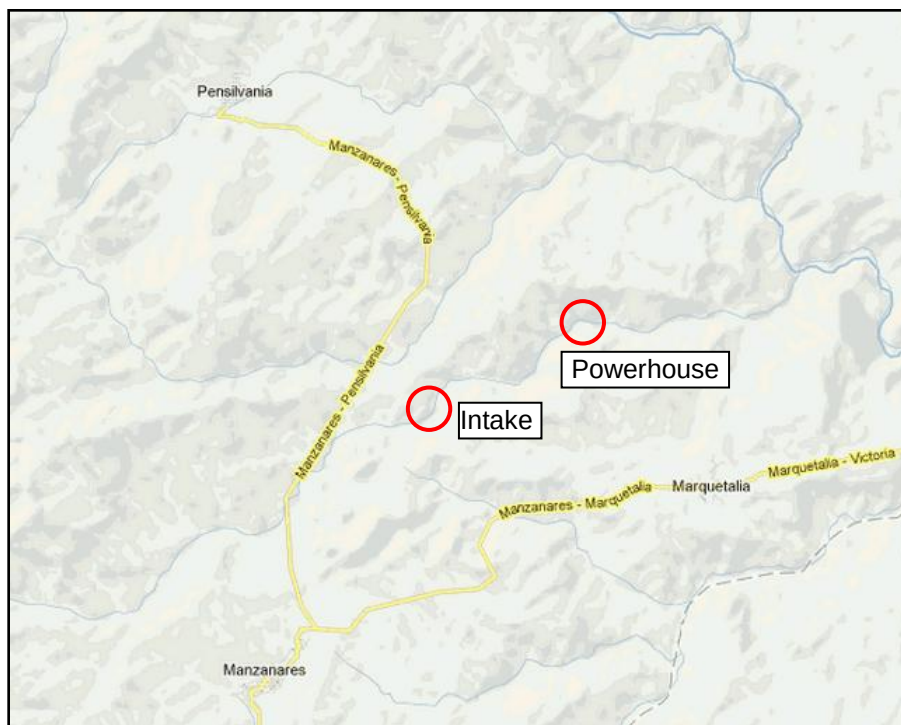


Figure 2: Project location in Colombia.

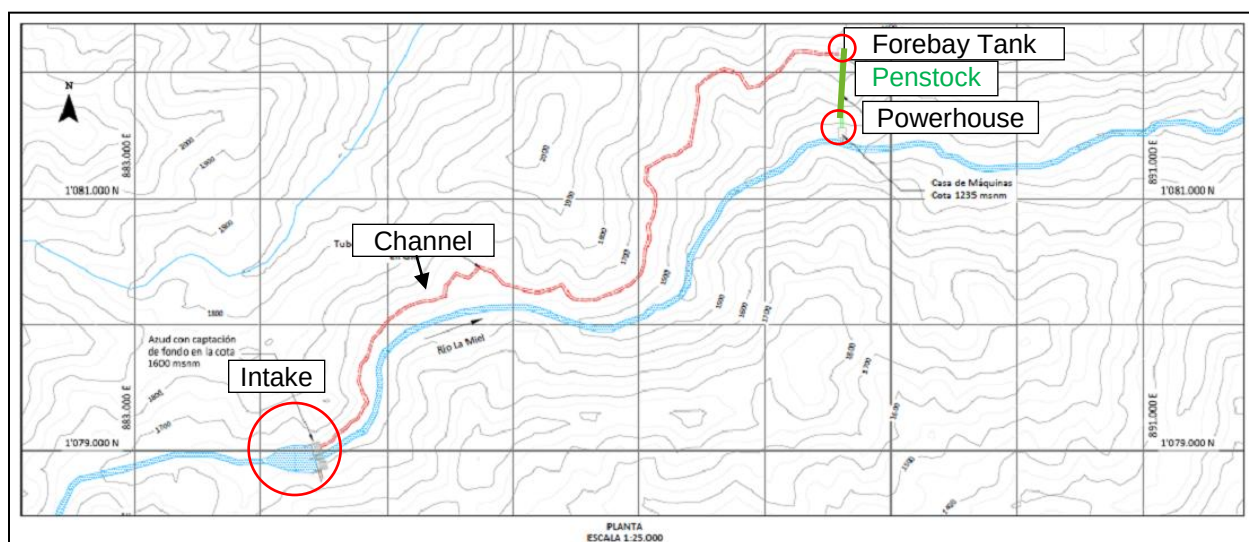


Figure 3: Project location.

1.10 Conditions Prior to Project Initiation

Prior to project implementation, no hydropower plant is installed at the site. The energy that will be delivered to the grid is generated by a mix of thermal and renewable power generation as reflected in the combined margin emissions factor (as per the tool applied). Since the project consists of a hydropower plant that does not generate GHG emissions itself, but reduces emissions of other power plants, it is clear

that it has not been implemented to generate GHG emissions for the purpose of their subsequent reduction, removal or destruction.

1.11 Compliance with Laws, Statutes and Other Regulatory Frameworks

The principal institutions responsible for Colombians Electricity Wholesale Market are⁴

- **The Ministry of Mines and Energy:** This is the leading institution in Colombia's energy sector.
- **The Unit for Mining and Energy Planning (UPME):** This unit of the Ministry of Mines and Energy is responsible for the study of future energy requirements and supply situations, as well as for drawing up the National Energy Plan and Expansion Plan.
- **The Regulatory Commission for Gas and Energy (CREG):** This entity is in charge of regulating the market for the efficient supply of energy. It defines tariff structures for consumers, transmission charges, and standards for the wholesale market, guaranteeing the quality and reliability of the service and economic efficiency. It also provides regulations that ensure the rights of consumers, the inclusion of environmental and socially sustainable principles, improved coverage and financial sustainability for participating entities.
- **XM Compañía de Expertos en Mercados S.A. E.S.P.:** This is a non-governmental agency acting as the market administrator, being in charge of the registration of contracts, the settlement and billing of all the transactions that take place in this market. XM is also in charge of the National Dispatch Center.

Table 3. Institutional structure of Colombian's electricity market⁴.

Policies	Ministry of Mines & Energy
Planning	Planning Unit of the Mines and Energy (UPME)
Regulation	Commission for the Regulation of Energy and Gas (CREG)
System Operation and Market Administration	XM Compañía de Expertos en Mercados S.A. E.S.P.

The structure of the Colombian energy market is based on Laws 142 (Public Services Law) and 143 (Electricity Law) of 1994⁵, which represent the last major reform of the power sector and establish the current regulatory framework. Since their enactment, Colombia has had a liberalized energy market, which is characterized by an unbundled generation, transmission, distribution, and commercialization scheme in order to separate the power activities and the markets. An electricity spot market and the development of a long-term contract market for electricity sales are the core of new structure to introduce a more effective framework for competition and an independent regulatory system supervised by the CREG (Regulatory Commission for Energy and Gas), created by the Law 143. This Electricity Law specifically introduced rules regarding: (i) Power sector planning; (ii) power generation; (iii) transmission and distribution; (iv) grid operation; (v) grid access fees; (vi) regime for electricity sales; (vii) concessions and contracts; and (viii) environmental issues, among others.

⁴ http://www.creg.gov.co/cxc/secciones/mercado_mayorista/estructura.htm, access date: 22/08/2011

⁵ Laws can be accessed on the website: <http://www.creg.gov.co/cxc/secciones/documentos/leyes.htm> (access date: 22/08/2011)

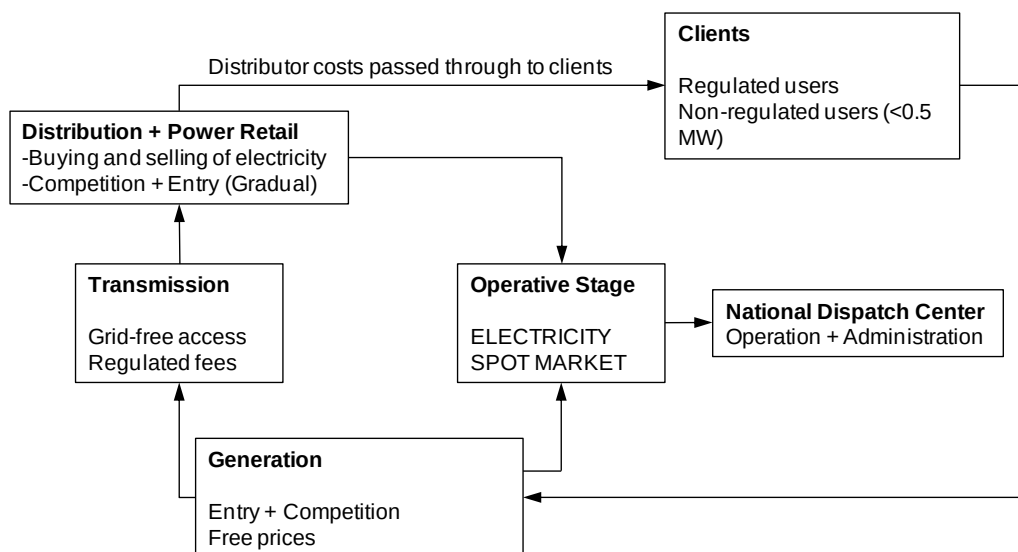


Figure 4: Simplified Scheme of the Colombian Power Market based on Electricity Law from 1994 (Law 143).

In order to construct and operate the hydroelectric plant and deliver electricity to the red, there are several instances that are directly involved to control that the project is implemented in accordance with the regulation and legal requirements as briefly explained in the following summaries:

1. **CORPOCALDAS:** Based on Article 49 of the law 99 of 1993 and article 3 of the decree 1220 of 2005 the project requires an environmental license to construct and operate the project. Since the project has less than 100 MW, based on Article 52 of law 99 and article 8 of decree 1220, the Regional Autonomous Corporation (in this case CORPOCALDAS) is responsible to control and supervise compliance of the project with the requirements established in the EIA and the environmental license. (For more information on the environmental license process, see section 5).
2. **UPME:** It acts as a supervisory entity. Any project to be implemented and interconnected with the electricity grid needs to be registered at the UPME. There are several phases, although the only mandatory registration step is called "Phase IV", which is emitted upon final project design and the emission of the environmental license. The project so far has obtained the registration "Phase I" and thus has a first approval of the UPME. The project will directly apply for registration "Phase IV" (expected for November 2013), without undergoing Phases II and III. By obtaining "Phase IV" it will be formally included in the power generation expansion plans of the UPME in its pipeline.
3. **XM Compañía de Expertos en Mercados S.A. E.S.P.:** Since the project will sell the energy under a power purchase agreement, it has to register it at XM, which is the grid administrator and operator. The entity that will buy the energy has also to be subscribed to XM and meet all applicable laws.
4. **CREG:** As the regulating entity, the CREG assures that all regulatory requirements are met. It also measures annually the capacity of the power plants, for instance to guarantee that minor plants are below the net generation capacity of 20 MW at the interconnection point.

Since the project is being constructed and operated in a highly regulated market that is controlled by a series of public and private actors, it has to be in compliance with all applicable laws and regulatory requirements.

1.12 Ownership and Other Programs

1.12.1 Right of Use

The proof of title is given by the environmental license assigned to Central Hidroeléctrica El Edén S.A.S E.P.S.

1.12.2 Emissions Trading Programs and Other Binding Limits

Not applicable.

1.12.3 Participation under Other GHG Programs

The project has not been registered and is not seeking registration under any other GHG programs.

1.12.4 Other Forms of Environmental Credit

The project has no intends to generate any other form of GHG-related environmental credit for GHG emission reductions or removals claimed under the VCS Program.

1.12.5 Projects Rejected by Other GHG Programs

The project has not has been rejected by any other GHG programs.

1.13 Additional Information Relevant to the Project

Eligibility Criteria

The guidelines of the form to complete the Project Document stated that “*For grouped projects, identify eligibility criteria for inclusion of new instances of each project activity.*” Since this is not a grouped project, this is not applicable to the project activity.

Leakage Management

Not applicable as per the applied methodology.

Commercially Sensitive Information

No information has been excluded.

Further Information

Not applicable.

2 APPLICATION OF METHODOLOGY

2.1 Title and Reference of Methodology

The project activity is developed in accordance with the approved consolidated CDM baseline and monitoring methodology ACM0002 “Consolidated baseline methodology for grid-connected electricity generation from renewable sources” (version 13.0.0).

(available at: <http://cdm.unfccc.int/methodologies/> , accessed: 15/03/2013)

The identification of the baseline scenario and the demonstration of additionality are assessed by applying the latest versions of the CDM “Tool for the demonstration and assessment of additionality” (version 7.0.0).

(available at: <http://cdm.unfccc.int/Reference/tools/index.html>, accessed: 15/03/2013)

The emission factor of the relevant power-grid is determined based on the procedures of the CDM “Tool to calculate the emission factor for an electricity system” (version 3.0.0).

Moreover, the following guidelines are applied:

“Guidelines on the assessment of investment analysis” (version 5) as per the Annex 5 of CDM EB 62 report.

“Guidelines on common practice” (version 02.0) as per the Annex 8 of CDM EB 69 report.

(available at: <http://cdm.unfccc.int/Reference/Guidclarif/index.html#pdd> , accessed: 15/03/2013)

2.2 Applicability of Methodology

The approved CDM methodology ACM0002 “Consolidated baseline methodology for grid-connected electricity generation from renewable sources” (version 13.0.0) is applicable to grid-connected renewable power generation project activities that:

- (a) install a new power plant at a site where no renewable power plant was operated prior to the implementation of the project activity (greenfield plant);
- (b) involve a capacity addition;
- (c) involve a retrofit of (an) existing plant(s); or
- (d) involve a replacement of (an) existing plant(s).

For this project, option (a) applies since a hydro power plant is installed at a site where no other renewable power plant operated before.

In the following it is explained how the individual applicability criteria from ACM0002 (version 13.0.0) are met:

- *The project activity is the installation, capacity addition, retrofit or replacement of a power plant/unit of one of the following types: hydro power plant/unit (either with a run-of-river reservoir or an accumulation reservoir), wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit;*

The proposed project activity consists of the installation and operation of a grid-connected hydro power plant.

- *In the case of capacity additions, retrofits or replacements (except for wind, solar, wave or tidal power capacity addition projects which use Option 2: on page 10 to calculate the parameter $EG_{PJ,y}$): the existing plant started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion or retrofit of the plant has been undertaken between the start of this minimum historical reference period and the implementation of the project activity.*

This condition is not relevant since the project activity does not involve any capacity additions, retrofit or replacements, but consists of the installation of a greenfield power plant.

- *In case of hydro power plants, one of the following conditions must apply:*
 - *The project activity is implemented in an existing single or multiple reservoirs, with no change in the volume of any of reservoirs; or*
 - *The project activity is implemented in an existing single or multiple reservoirs, where the volume of any of reservoirs is increased and the power density of each reservoir, as per the definitions given in the project emissions section, is greater than 4 W/m²; or*
 - *The project activity results in new single or multiple reservoirs and the power density of each reservoir, as per the definitions given in the project emissions section, is greater than 4 W/m².*

The project activity is a hydro power plant without reservoir; thus this criterion is not relevant.

- *In case of hydro power plants using multiple reservoirs where the power density of any of the reservoirs is lower than 4 W/m² all the following conditions must apply:*
 - *The power density calculated for the entire project activity using equation 5 is greater than 4 W/m²;*
 - *Multiple reservoirs and hydro power plants located at the same river and where are designed together to function as an integrated project⁶ that collectively constitute the generation capacity of the combined power plant;*
 - *Water flow between multiple reservoirs is not used by any other hydropower unit which is not a part of the project activity;*
 - *Total installed capacity of the power units, which are driven using water from the reservoirs with power density lower than 4 W/m², is lower than 15 MW;*
 - *Total installed capacity of the power units, which are driven using water from reservoirs with power density lower than 4 W/m², is less than 10% of the total installed capacity of the project activity from multiple reservoirs.*
- The project activity is a hydro power plant without reservoir; thus this criterion is not relevant.

The methodology is not applicable to the following:

⁶ This requirement can be demonstrated, for example: (i) by the fact that water flow from upstream power units spilling directly to the downstream reservoir; or (ii) through the analysis of the water balance. Water balance is the mass balance of water fed to power units, with all possible combinations of multiple reservoirs and without the construction of reservoirs. The purpose of such water balance is to demonstrate the requirement of specific combination of multiple reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum three years prior to implementation of CDM project activity.

- *Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site;*

The proposed project activity does not involve switching from fossil fuels to renewable energy at the site of the project activity.

- *Biomass fired power plants;*

The proposed project activity does not involve biomass fired power plants.

- *Hydro power plants that result in new reservoirs or in the increase in existing reservoirs where the power density of the power plant is less than 4 W/m².*

The project activity is a hydro power plant without reservoir; thus this criterion is not relevant.

- *In the case of retrofits, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is "the continuation of the current situation, i.e. to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance".*

This condition is not relevant since the project activity does not involve any capacity additions, retrofit or replacements, but consists of the installation of a completely new power plant.

Thus, the methodology is applicable to the proposed project activity.

The "Tool to calculate the emission factor for an electricity system" (version 3.0.0) is applied to calculate baseline emissions for a project activity that substitutes grid electricity. Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, there are specific conditions that should be met. Since the electricity system affected by the proposed project activity includes only grid connected power plants, no specific conditions has to be assessed. Further, the tool would not be applicable if the project electricity system is located partially or totally in an Annex-I country, which for the proposed project activity is not the case.

2.3 Project Boundary

According to the guidance specified in the Methodology ACM0002 (version 13.0.0), the spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system that the project activity is connected to (see Figure 5). The measuring equipment is located at the interconnection point of the project with the SIN, which is the substation.

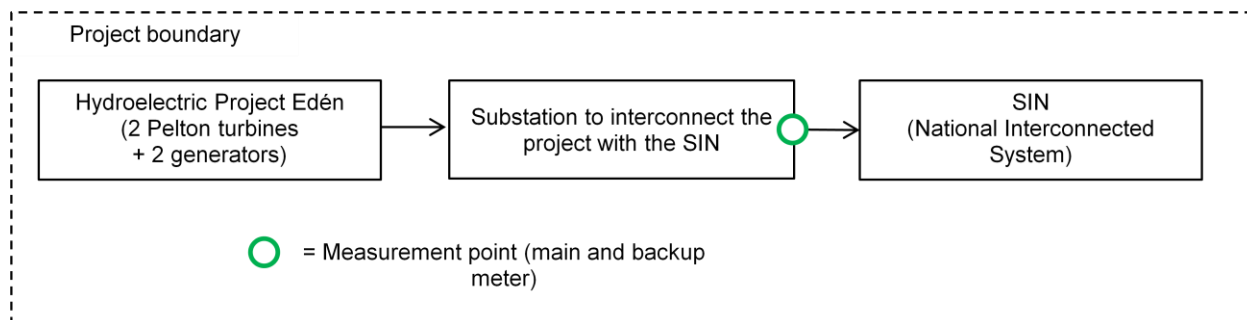


Figure 5. Project boundary and measurement point.

The spatial extent of the project boundary is the National Interconnected System (SIN⁷) of Colombia. The power plants of this grid are all connected and can be dispatched without significant transmission constraints.

Source		Gas	Included?	Justification/Explanation
Baseline	Electricity generation in fossil fuel fired power plants that are displaced due to the project activity	CO ₂	Yes	According to ACM0002, major emission source of emissions in the baseline
		CH ₄	No	According to ACM0002, minor emission source, excluded for simplification.
		N ₂ O	No	According to ACM0002, minor emission source, excluded for simplification.
		Other	No	No other relevant emissions in the baseline exist. This is conservative.
Project	Emissions of CH ₄ from the reservoir	CO ₂	No	According to ACM0002, minor emission source
		CH ₄	No	According to ACM0002, since power density is 63 W/m ² , larger than 10 W/m ² , this can be neglected
		N ₂ O	No	According to ACM0002, minor emission source
		Other	No	No other relevant emissions from the project occur.

2.4 Baseline Scenario

As stated in the approved methodology ACM0002 “Consolidated baseline methodology for grid-connected electricity generation from renewable sources”, version 13.0.0: If the project activity is the installation of a new grid-connected renewable power plant/unit, the baseline scenario is the following:

Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the “Tool to calculate the emission factor for an electricity system”.

Therefore, the baseline scenario consists of the electricity that would have been generated and delivered to the grid in the absence of the proposed project activity by:

- Other plants currently connected to the SIN; and
- New capacity additions to the SIN.

Hence, the baseline scenario is identified as the continuation of the common practice of power generation, i.e. mainly large hydro power plants with reservoirs and thermal power stations that emit large quantities of carbon dioxide (CO₂) to the atmosphere.

⁷ in Spanish: *Sistema Interconectado Nacional*

2.5 Additionality

The timeline of project implementation can be summarized as following:

Table 4. Overview of key events in the development of the proposed project

Date	Key event
until Jan 2010	Project identification
Jan-Dic 2010	Feasibility and environmental studies
Nov-Dic 2010	Environmental Impact Assessment (EIA) submitted to Corpocaldas (Autonomous Corporation of Caldas ⁸) for requesting the Environmental License.
May 2011	Issuance of Environmental License (Resolution 173 of 4 th of May 2011) for construction and operation of the hydroelectric plant El Edén. The license was issued to Latinoamericana de Construcciones S.A. (Latinco S.A.) and later assigned to the project proponent.
July 2011	Constitution of the company Central Hidroeléctrica El Edén S.A.S E.S.P on July 7, 2011, to promote, develop and execute the project activity, including also the administration, operation and maintenance of the plant, as well as the energy generation and sales.
September 2011	The transfer of the environmental license from Latinoamericana de Construcciones S.A. to Central Hidroeléctrica El Edén S.A.S E.S.P is approved by Corpocaldas on September 22, 2011.
April 2012	The construction contract is signed between Unión Temporal Grupo Constructor Centrales Hidroeléctricas del Oriente de Caldas (UT GS CHOC) on April 16, 2012.
May 2012	Reception of proposal for energy sales agreement on May 10, 2012, from the company Synermin. Upon the proposal for the energy sales agreement, the investment decision was made in order to go ahead with project implementation. The presentation was made on May 16, 2012. This is the date of investment decision making.
November 2012	The energy sales agreement is signed on November 16, 2012 between Central Hidroeléctrica El Edén S.A.S E.S.P and Generarco S.A. E.S.P, which is from the same group as Synermin (the company that made the proposal). The conditions of the energy sales agreement are the same as in the proposal.
June 2013	Expected construction start
January 2015	The project is expected to start operation on January 1, 2015.

The additionality of the project activity is demonstrated and assessed applying the CDM “Tool for the demonstration and assessment of additionality” (version 7.0.0), as stated in the methodology ACM0002 (version 13.0.0) and accepted by the VCS Standard 3 (version 3.3).

The tool provides a step-wise approach to demonstrate and assess additionality:

- Step 0. Demonstration whether the proposed project activity is the first-of-its-kind;
- Step 1. Identification of alternatives to the project activity;
- Step 2. Investment analysis to determine that the proposed project activity is either: (1) not the most economically or financially attractive, or (2) not economically or financially feasible;
- Step 3. Barriers analysis; and
- Step 4. Common practice analysis.

⁸ in Spanish: *Corporación Autónoma de Caldas*

Step 0: Demonstration whether the proposed project activity is the first-of-its-kind

This step is optional. It is not applied for this project activity and therefore it is considered that the proposed project activity is not the first-of-its-kind.

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations

Realistic and credible alternatives to the project activity(s) are defined through the following sub-steps:

Sub-step 1a: Define alternatives to the project activity

According to the Tool for the demonstration and assessment of additionality, project activities applying this tool in context of approved consolidated methodology ACM0002, only need to identify that there is at least one credible and feasible alternative that would be more attractive than the proposed project activity.

Outcome of Step 1a: For the project proponent, the possible alternatives to the proposed project include:

Alternative 1: The proposed project activity undertaken without being registered as a CDM project activity.

Alternative 2: Continuation of the current situation: In this case, the project activity will not be constructed and the power will be solely supplied by the operation of power plants connected to the SIN and by the addition of new power plants.

Sub-step 1b: Consistency with mandatory laws and regulations

The implementation and operation of the project activity affects two different groups of power plants in the interconnected national system as represented in the baseline scenario by the combined margin:

- Existing power plants whose current electricity generation would be affected by the proposed CDM project activity; and
- Prospective power plants whose construction and future operation would be affected by the proposed CDM project activity.

Both alternatives, i.e. the proposed project activity undertaken without being registered as a CDM project and the continuation of the current situation, are in compliance with all of the mandatory laws and regulations as described in section 1.11.

In the following, it is shown that the proposed project is additional as per the latest guidelines of the VCS Standard 3 (version 3.3). In order to assess additionality, the corresponding CDM tool is applied as explained step by step.

Outcome of Step 1b: Both alternatives identified in sub-step 1a are in compliance with mandatory laws and regulations:

Alternative 1: The proposed project activity undertaken without being registered as a CDM project activity.

Alternative 2: Continuation of the current situation: In this case, the project activity will not be constructed and the power will be solely supplied by the operation of power plants connected to the SIN and by the addition of new power plants.

Step 2: Investment analysis

To conduct the investment analysis, following Sub-steps are used:

Sub-step 2a: Determine appropriate analysis method

According to CDM “Tool for the demonstration and assessment of additionality” (version 7.0.0), three options can be applied for the investment analysis: the simple cost analysis, the investment comparison analysis and the benchmark analysis.

The project activity generates financial and economic benefits other than CER revenues, therefore the simple cost analysis (Option I) is not applicable. The corresponding CDM “Guidelines on the assessment of investment analysis” (version 5), states that: *“if the alternative to the project activity is the supply of electricity from a grid this is not to be considered an investment and the benchmark approach is considered appropriate”*. Thus, Option II (investment comparison analysis) is also not applicable and the benchmark analysis (Option III) is chosen to prove additionality.

Sub-step 2b – Option III: Apply benchmark analysis

The additionality tool requires an identification of the most appropriate financial indicator. For the case of a power plant that would supply energy to the grid, the most appropriate indicator is the internal rate of return (IRR) as it characterizes the rate of return on invested capital. In this analysis an equity IRR is calculated in accordance with the additionality tool and the corresponding guidelines as indicated above. Taxation is included as an expense in the IRR calculation, i.e. the IRR is determined as a post-tax indicator.

In accordance with the “Guidelines on the assessment of investment analysis” (version 5) a default value for the expected return on equity is used for the benchmark. The relevant benchmark for energy projects in Colombia (Group 1 as given in the guidelines) is 12.00% in real terms. As per the guidelines, since the investment analysis is carried out in nominal terms, the real term values provided can be converted to nominal values by adding the inflation rate. Since no long-term inflation forecasts or target rates of the central bank for the duration of the crediting period exist, the average forecasted inflation rate of 2.90% for the next five years published by the IMF (International Monetary Fund World Economic Outlook) is used (based on the forecasts for the period from 2012 to 2016, consistent with the start of the VCS validation and the data available at the IMF until maximum 2016).

The benchmark, i.e. the Nominal Return on Equity, is therefore given as 14.90%.

Sub-step 2c: Calculation and comparison of financial indicators

For the financial analysis the main cash outflows are given by the investment, the ongoing O&M costs and taxes. The cash inflows are mainly generated from revenues of electricity sales.

Input values for the investment analysis

The electricity sales are made under an energy sales agreement with stepwise prices, depending on the energy generation.

The financial structure is applied as suggested by the guidelines for investment analysis (version 5). For the equity/debt structure, the default ratio of 50% equity and 50% debt is used. The debt terms are taken from a proposal of a financial institution of Colombia⁹. A summary of the debt terms used for the investment analysis is given in the following table.

⁹ The name is kept confidential for the public. The proposal is submitted to the DOE.

Table 5: Terms of debt structure¹⁰.

Parameter	Value	Source
Total investment (USD)	\$48,284,307	Construction contract. ¹¹ The value in the contract is 85,000,000 COP in real terms as on December 2011. This is annually incremented as per the Colombian CPI and converted to US Dollars (for details of increments and conversion see financial analysis spreadsheets "Investment" and "Financial Indicators").
% Debt	50%	Based on the default value of the Guidelines for Investment Analysis
Equity amount (USD)	\$24,142,154	Calculated (in nominal terms)
Loan amount (USD)	\$24,142,154	Calculated (in nominal terms)
Interest rate (p.a.)	11.20%	The average of the most recent three years prior to decision making is taken (2009-2011). ¹²
Principal (annual)	\$2,414,215	Annual principal during 10 years

Table 6 lists the parameters and values used for carrying out the investment analysis. All input values are valid at investment decision on May 16, 2012 (see beginning of this section). All sources and further details are provided in the Investment Analysis spreadsheets.

An assessment period of 33 years is applied, accounting for the estimated construction period of 3 years of construction and the maximum of 30 years of operation under the renewable crediting scheme of the VCS. Since the project has an expected lifetime of 50 years, a fair value is included in the last year of the assessment period as per the Annex 8 "Clarification: Applicability of the "Guidelines on the assessment of investment analysis" (version 01.0)" of the UNFCCC EB73 report.¹³

The fair value in the last year of assessment (year 30 of operation) is calculated by applying the discounted cash flow (DCF) method¹⁴ for the remaining years of operation (years 31 to 50). The cash flows of those remaining years of project operation are calculated in the same manner as the cash flows in the years before by extrapolating the individual projections. They are discounted to the year 30 and included as a fair value. For the discount rate the commercial lending rates are chosen as per option (b) of paragraph 38 of the tool. (For details of sources and parameters, see financial spreadsheet).

¹⁰ For details and specific sources, see spreadsheet "Debt" of the Investment Analysis in the Excel file "Equity IRR Hydroelectric Project El Edén"

¹¹ Cto de Construcción_UT GC CHOC (EPC).pdf

¹² <http://www.tradingeconomics.com/colombia/lending-interest-rate-percent-wb-data.html>

¹³ The valuation period of the CDM is shorter since the renewal of the crediting period is also limited to 21 years. Thus the period is extrapolated to be more representative for the VCS crediting period.

¹⁴ This is basically the same as calculating a net present value (NPV). A brief explanation of the method is given at: <http://www.investopedia.com/terms/d/dcf.asp>

Sources that indicate the DCF method as a valid approach are for example:

<https://inform.pwc.com/inform2/show?action=informContent&id=1318132703110207>

<http://www.cica.ca/applying-the-standards/financial-reporting/accounting-standards-for-private-enterprises/item52857.pdf> (page 2)

Table 6. Input values used in the investment analysis available at the moment of decision making (all sources and calculations are provided in the Investment Analysis spreadsheets¹⁵).

Construction starts	1-Jul-13	
Construction period	30	months
Operational start	1-Jan-15	
<i>Electricity generation</i>		
Total net energy generation for sales	104.5	GWh / year
Firm energy (tariff I)	43.5	GWh / year
Surplus energy (tariff II)	61.0	GWh / year
Tariffs		
<i>Electricity sales</i>		
Energy tariff I (up to 43.5 GWh/yr)	132.00	COP/kWh
Energy tariff II (surplus energy)	105.00	COP/kWh
INVESTMENT		
<i>Total Capital Costs</i>		
Total Investment	\$48,284,307	USD
Equipment cost (for depreciation only)	\$3,954,333	USD
OPERATING COSTS & EXPENSES		
<i>Operation costs</i>		
Expenses (this includes operation, maintenance, administration and commercial costs)	\$641,070	USD/year
<i>Taxes</i>		
Income taxes	33.0%	%
<i>Depreciation</i>		
Annual depreciation Civil Works + Infrastructure	20	years
Annual depreciation equipment	10	years
FINANCIAL PARAMETERS		
<i>Macro</i>		
Exchange rate	1,807	COP/USD
Producer Price Index (PPI)	2.76%	%
Consumer Price Index (CPI)	2.92%	%
Inflation rate (forecast IMF)	2.90%	%
Discount rate	11.20%	%
<i>Benchmark</i>		
Return on Equity (real terms)	12.00%	%
Inflation Adjustment	2.90%	%
BENCHMARK: Nominal Return on Equity	14.90%	%

Result of the investment analysis

Based on the parameters above, the Internal Rate of Return (post-tax equity IRR) is calculated as 12.22%, which is below the benchmark rate of 14.90%.

¹⁵ see "Equity IRR Hydroelectric Project El Edén"

Sub-step 2d: Sensitivity analysis

A sensitivity analysis is carried out by varying the following key parameters to analyze the impact on the equity IRR:

- Energy generation (MWh): The quantity of energy generated is varied.
- Investment costs (USD): the complete investment cost is varied.
- O&M costs (USD/year): the complete O&M cost is varied.

The energy tariff is not varied since it will be fixed under the energy sales agreement.

Table 7 shows a sensitive analysis for the resulting IRR by varying the main parameters +/- 10%.

Table 7: Sensitivity analysis.

Variation of electricity generation	Base	+10.00%
IRR	12.22%	13.21%

Variation of electricity tariff	Base	+10.00%
IRR	12.22%	13.32%

Variation of investment costs	Base	-10.00%
IRR	12.22%	13.33%

Variation of O&M costs	Base	-10.00%
IRR	12.22%	12.32%

As can be seen, the benchmark is not reached in these scenarios; thus it can be concluded that the project activity is not financially attractive.

Step 3: Barrier analysis

The project activity does not apply a barrier analysis.

Step 4: Common practice analysis

For the assessment of common practice analysis, the “Guidelines on Common Practice” (version 02.0) provided in Annex 8 of the EB Report 62 are applied. The applicable geographical area for the assessment of common practice analysis corresponds to the host country Colombia.

According to the guidelines for common practice analysis, the following steps are applied¹⁶:

Step 1: Calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity.

For the common practice analysis, the installed capacity of 20.23 MW is used, and therefore the applicable range as +/-50% of the capacity is 10.115 MW to 30.345 MW.

Step 2: Identify similar projects (both CDM and non-CDM).

¹⁶ For the detailed analysis, see Excel file “Common Practice Analysis El Edén”

The information about all operational power plants connected to the National Interconnected System (SIN) is obtained from XM (*Expertos en Mercados*), which is the grid operator and administrator of the SIN. The following power plants have a power generation capacity within the capacity range identified in step 1:

Table 8: Power plants in the same capacity range (+/-50%).

PLANT	CAPACITY [MW]	RESOURCE	CDM
BARROSO	19.90	Water	x
HIDROMONTAÑITAS	19.90	Water	x
INSULA	19.00	Water	
MCALDERAS	19.90	Water	x
MENOR ALTO TULUA	19.90	Water	x
MENOR AMAIME	19.90	Water	x
MFLORIDA	19.90	Water	
MRIO MAYO	19.80	Water	
RIO PIEDRAS	19.90	Water	
LA HERRADURA	19.80	Water	x
LA JUNCA	19.40	Water	
LA TINTA	19.40	Water	
LA VUELTA	11.60	Water	x
MENOR RIOGRANDE	19.00	Water	
NIQUIA	19.00	Water	
PALMAS SAN GIL	15.00	Water	
SAN ANTONIO	19.40	Water	
SONSON	18.50	Water	
TEQUENDAMA	19.40	Water	
AYURA	18.00	Water	
CHARQUITO	19.40	Water	
EL LIMONAR	18.00	Water	
CIMARRON	19.90	Gas	
EL MORRO I	19.90	Gas	
EL MORRO II	19.90	Gas	
TERMOYOPAL	19.90	Gas	
JEPIRACHI	18.42	Wind	x
COGENERADOR MAYAGUEZ	19.90	Coal+Bagasse	
COGENERADOR INGENIO PROVIDENCIA	19.90	Coal+Bagasse	
TERMOYOPAL 2	30.00	Gas	
ESMERALDA	30.0	Water	

greyed out = CDM projects

Step 3: Within the projects identified in Step 2, identify those that are neither registered CDM project activities, project activities submitted for registration, nor project activities undergoing validation. Note their number Nall.

From the total of 31 power plants in the same capacity range (step 2), there are 8 projects submitted for registration or undergoing validation; thus $N_{all} = 23$. (i.e. all projects in Table 8 that are not greyed out)

Step 4: Within plants identified in Step 3, identify those that apply technologies that are different to the technology applied in the proposed project activity. Note their number as N_{diff} .

As per the guidelines, different technologies are technologies that deliver the same output and differ by at least one of the following (as appropriate in the context of the measure applied in the proposed CDM project and applicable geographical area):

- (i) Energy source/fuel (example: energy generation by different energy sources such as wind and hydro and different types of fuels such as biomass and natural gas);
- (ii) Feed stock (example: production of fuel ethanol from different feed stocks such as sugar cane and starch, production of cement with varying percentage of alternative fuels or less carbon-intensive fuels);
- (iii) Size of installation (power capacity) / energy savings:
 - Micro (as defined in paragraph 24 of decision 2/CMP.5 and paragraph 39 of decision 3/CMP.6);
 - (Small (as defined in paragraph 28 of decision 1/CMP.2);
 - Large.
- (iv) Investment climate in the date of the investment decision, inter alia:
 - Access to technology;
 - Subsidies or other financial flows;
 - Promotional policies;
 - Legal regulations;
- (v) Other features, inter alia:
 - Nature of the investment (example: unit cost of capacity or output is considered different if the costs differ by at least 20 %).

In the following it is shown why most of the 23 non-CDM power plants in the same capacity range are different from the proposed project activity.

An important distinction can be made by the “Energy source/fuel”, since there are other technologies involved, using other resources (thermal power and cogeneration). The following seven plants have a different energy source and different energy conversion technologies:

Table 9: Thermal Power plants using different resources and technologies.

POWER PLANT	MW	RESOURCE
CIMARRON	19.90	Gas
EL MORRO I	19.90	Gas
EL MORRO II	19.90	Gas
TERMOYOPAL 1	19.90	Gas
COGENERADOR MAYAGUEZ	19.90	Coal+Bagasse
COGENERADOR INGENIO PROVIDENCIA	19.90	Coal+Bagasse
TERMOYOPAL 2	30.00	Gas

Further, as explained in section 1.11, the power sector underwent a major reform in year 1994, when the laws 142 and 143 were enacted, which define the current market structure and regulatory framework. Power plants implemented before that year had a completely different legal and regulatory framework;

and therefore, as per item (iv) above, are considered “different technologies” than in the proposed project activity. The following 15 projects started operation before the new energy law:

Table 10: Hydroelectric projects constructed before.

POWER PLANT	OPERATIONAL START
INSULA	1951
MFLORIDA	1975
MRIO MAYO	1969
LA JUNCA	1962
LA TINTA	1967
MENOR RIOGRANDE	1956
NIQUIA	1993
PALMAS SAN GIL	1954
SAN ANTONIO	1963
SONSON	1965
TEQUENDAMA	before 1985 (exact year not published)
AYURA	1983
CHARQUITO	1972
EL LIMONAR	1957
ESMERALDA	1963

Based on these criteria, the total number of different plants is $N_{diff} = 22$.

Step 4: Calculate factor $F=1-N_{diff}/N_{all}$ representing the share of plants using technology similar to the technology used in the proposed project activity in all plants that deliver the same output or capacity as the proposed project activity.

As per the guidelines, a project is considered a common practice within a sector in the applicable geographical area if the factor F is greater than 0.2 and $N_{all} - N_{diff}$ is greater than 3. Since $N_{all} = 23$ and $N_{diff} = 22$, $N_{all} - N_{diff} = 1$ and

— —

Therefore the project is clearly not a common practice.

Conclusion of the additionality analysis

Since the project activity is not financially attractive (step 2) and the common practice analysis shows that it is not business-as-usual (Step 4), the proposed project activity is additional.

2.6 Methodology Deviations

Not applicable. No methodology deviations are made.

3 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

3.1 Baseline Emissions

The baseline scenario and the emission rate calculation are based on the electricity that otherwise would have been generated by the plants connected to the grid and by addition of future plants. According to the methodology, no leakage emissions are considered.

Baseline emissions are calculated by multiplying the combined margin emission factor ($EF_{grid,y}$, in tCO_2e/MWh) by the electricity generated by the proposed project activity during the year y (EG_y , in MWh).

Baseline emissions include only CO_2 emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation by the project activity would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants.

The detailed calculations and data of the baseline emissions are presented in the Excel file “EF Hydroelectric Project El Edén”. In the following, the relevant methodological approaches and formulas are presented.

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y} \quad (1)$$

Where:

- BE_y = Baseline emissions in year y (tCO_2/yr)
- $EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)
- $EF_{grid,y}$ = Combined margin CO_2 emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (tCO_2/MWh)

Since it is a green-field project, according to Eq. (7) of ACM0002,

$$EG_{PJ,y} = EG_{facility,y}$$

The expression $EG_{PJ,y}$ is used in this PDD.

The combined margin emission factor ($EF_{grid,CM,y}$) is calculated following the guidance in the “Tool to calculate the emission factor for an electricity system” (version 3.0.0) by applying the following steps:

- STEP 1. Identify the relevant electricity systems;
- STEP 2. Choose whether to include off-grid power plants in the project electricity system (optional);
- STEP 3. Select a method to determine the operating margin (OM);
- STEP 4. Calculate the operating margin emission factor according to the selected method;
- STEP 5. Calculate the build margin (BM) emission factor;
- STEP 6. Calculate the combined margin (CM) emissions factor.

In the following it is explained how each step is applied.

STEP 1. Identify the relevant electricity systems.

For determining the electricity emission factors, the project electricity system is defined by the spatial extent of the power plants that are physically connected through transmission and distribution lines to the

project activity (i.e., Hydroelectric Project El Edén) and that can be dispatched without significant transmission constraints. In this case, the project electricity system is given as the National Interconnected System (SIN).

For the purpose of determining the operating margin emission factor, the CO₂ emission factor(s) for net electricity imports is chosen as 0 t CO₂/MWh.

STEP 2. Choose whether to include off-grid power plants in the project electricity system (optional).

In accordance with the tool, this step is optional. For the proposed project activity, off-grid power plants are not included in the project electricity system (Option 1).

STEP 3. Select a method to determine the operating margin (OM).

In accordance with the tool, the calculation of the operating margin emission factor (EF_{grid,OM,y}) is based on one of the following methods:

- (a) Simple OM; or
- (b) Simple adjusted OM; or
- (c) Dispatch data analysis OM; or
- (d) Average OM.

For the project activity, the simple adjusted OM is applied, using the *ex ante* data vintage:

Ex ante option: if the ex ante option is chosen, the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the emissions factor during the crediting period is required. For grid power plants, use a 3-year generation-weighted average, based on the most recent data available at the time of submission of the VCS-PD for validation. For off-grid power plants, use a single calendar year within the five most recent calendar years prior to the time of submission of the VCS-PD for validation.¹⁷

All power plants connected to the SIN are included. Power plants registered as CDM project activities are also included as suggested by the tool. Historical data of the three most recent years is available from XM (grid operator and administrator¹⁸).

STEP 4. Calculate the operating margin emission factor according to the selected method.

The simple adjusted operating margin emission factor EF_{grid,OM-adj,y} (tCO₂e/MWh) is a variation of the simple operating margin emission factor, where the power sources (including imports) are separated in low-cost/must-run power sources (k) and other power sources (j), as follows:

$$EF_{grid,OM-adj,y} = (1 - \lambda_y) \cdot \frac{\sum_m EG_{m,y} \cdot EF_{EL,m,y}}{\sum_m EG_{m,y}} + \lambda_y \cdot \frac{\sum_{i,j} EG_{k,y} \cdot EF_{EL,k,y}}{\sum_k EG_{k,y}} \quad (2)$$

Where:

- λ_y = Factor expressing the percentage of time when low-cost/must-run power units are on the margin in year y
- $EG_{m,y}$ = Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)

¹⁷ The term “CDM-PDD” given in the original text was replaced by “VCS PD” for this project activity. The CDM term “DOE” has also been removed.

¹⁸ www.xm.com.co

$EG_{k,y}$	= Net quantity of electricity generated and delivered to the grid by power unit k in year y (MWh)
$EF_{EL,m,y}$	= CO ₂ emission factor of power unit m in year y (tCO ₂ /MWh)
$EF_{EL,k,y}$	= CO ₂ emission factor of power unit k in year y (tCO ₂ /MWh)
m	= All grid power units serving the grid in year y except low-cost/must-run power units
k	= All low-cost/must run grid power units serving the grid in year y

The lambda factor (λ) is determined as:

$$\lambda = \frac{\text{number of hours per year low – cost / must – run sources are on the margin}}{8760 \text{ hours per year}} \quad (3)$$

According to the methodology, the number of hours low-cost/must-run sources are on the margin is obtained through the following procedure (see Figure 6 below):

Step i) Plot a Load Duration Curve

Collect chronological load data (typically in MW) for each hour of the year y , and sort the load data from the highest to the lowest MW level. Plot MW against 8760 hours in the year in descending order.

Step ii) Organize Data by Generating Sources

Collect electricity generation data from each power plant/unit. Calculate the total annual generation (in MWh) from low-cost/must-run power plants/units.

Step iii) Fill Load Duration Curve

Fill the load duration curve. Plot a horizontal line across the load duration curve such that the area under horizontal line and the curve right from the intersection point (MW times hours) equals the total generation (in MWh) from low-cost/must-run power plants/units

Step iv) Determine the “Number of hours per year low-cost/must-run sources are on the margin”

Determine the “Number of hours for which low-cost/must-run sources are on the margin in year y . First, locate the intersection of the horizontal line plotted in Step (iii) and the load duration curve plotted in Step (i). The number of hours (out of the total of 8760 hours) to the right of the intersection is the number of hours for which low-cost/must-run sources are on the margin. If the lines do not intersect, then one may conclude that low-cost/must-run sources do not appear on the margin and lambda is equal to zero.

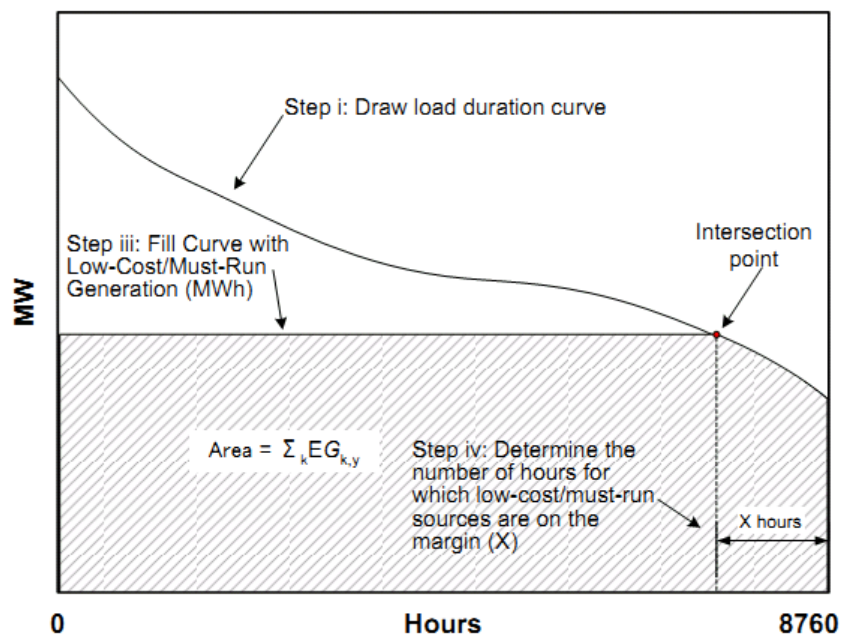


Figure 6: Illustration of Lambda calculation for simple adjusted OM method. (Note: Step (ii) is not shown in the figure; it deals with organizing data by source.)

The detailed calculations of lambda are provided in the Excel file “EF Colombia 2010-2012”. The following figures show the load duration curves and the area given by low cost/must run units, and the resulting lambda.

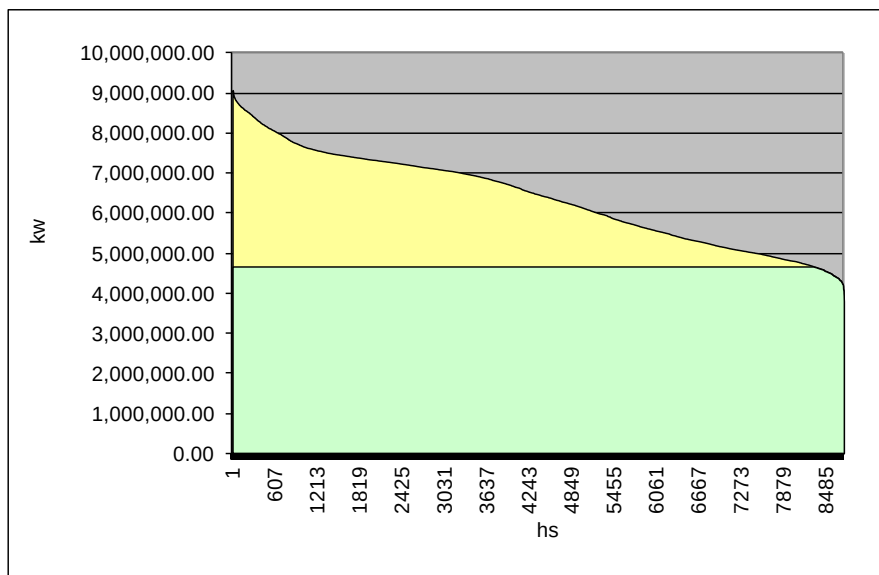


Figure 7: Load duration curve and area given by low-cost/must-run units, and the resulting lambda in year 2009.

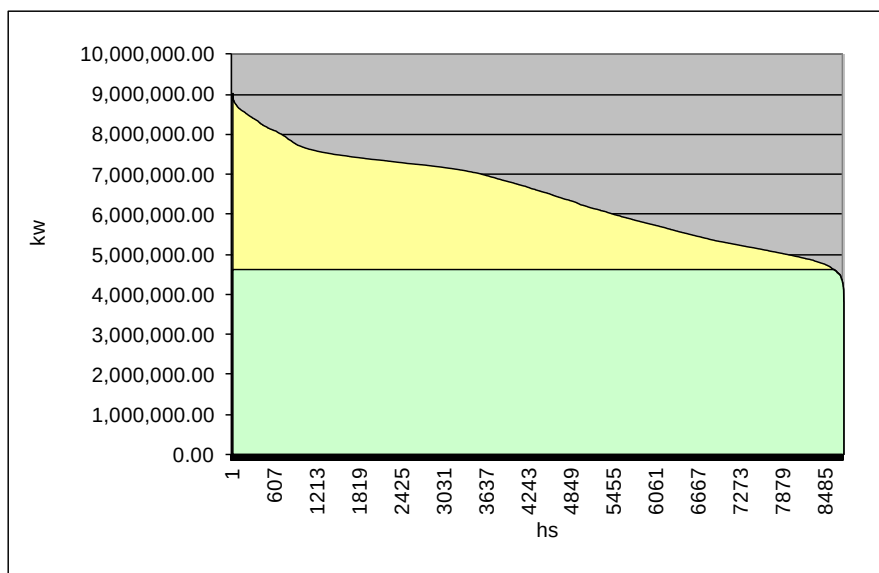


Figure 8: Load duration curve and area given by low-cost/must-run units, and the resulting lambda in year 2010.

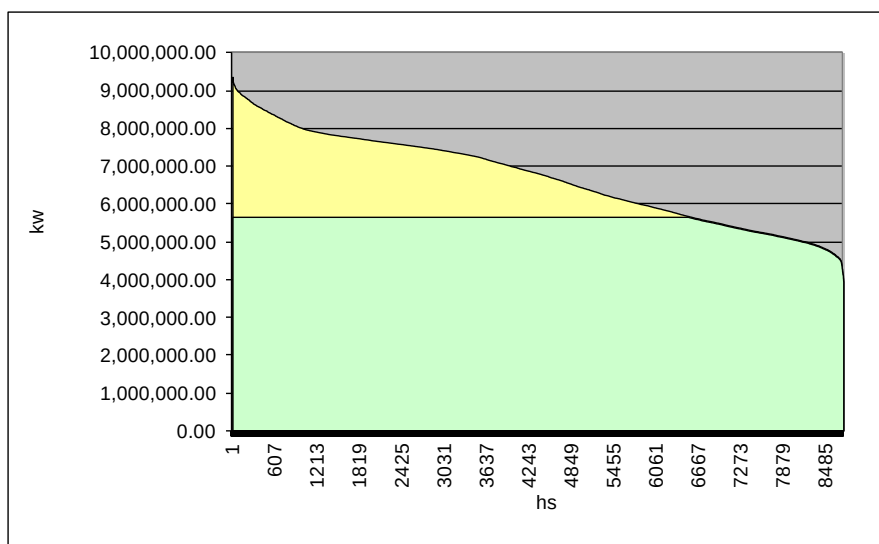


Figure 9: Load duration curve and area given by low-cost/must-run units, and the resulting lambda in year 2011.

Determination of $EF_{EL,m,y}$

The emission factor of each power unit m is determined as follows (power units k are not included, since the low-cost/must-run units have zero emissions and thus do not require calculating the emission factor):

For the simple adjusted OM “Option A: Calculation based on average efficiency and electricity generation of each plant” of the tool is chosen as the most appropriate approach. Option A2 is the appropriate method due to the availability of heat rates (efficiencies) of each plant:

$$EF_{EL,m,y} = \frac{EF_{CO_2,m,i,y} \times 3.6}{\eta_{m,y}} \quad (4)$$

Where:

- $EF_{EL,m,y}$ = CO₂ emission factor of power unit m in year y (tCO₂/MWh)
 $EF_{CO_2,m,i,y}$ = Average CO₂ emission factor of fuel type i used in power unit m in year y (t CO₂/GJ)
 $\eta_{m,y}$ = Average net energy conversion efficiency of power unit m in year y (ratio)

By applying formulae (4) to determine the emission factor of each power plant, the results from the lambda calculation and the main equation (2) for the OM emission factor, and the corresponding generation weights of each year, the simple OM emission factor is determined as shown in Table 11.

Table 11: Simple adjusted operating margin emission factor of the years 2010-2012.

	2010	2011	2012
EF No LC/MR	0.5797	0.5495	0.5597
EF LC/MR	0.0000	0.0000	0.0000
Lambda	0.0174	0.2547	0.1717
EF OM [tCO ₂ /MWh]	0.5696	0.40957	0.46363
Generation [MWh]	56,798,608	58,615,075	59,995,370
EF OM Simple adjusted 09,10,11 (tCO₂/MWh)	0.4798		

STEP 5. Calculate the build margin (BM) emission factor

In terms of vintage of data, option 1 of the tool is chosen, i.e. the *ex ante* approach:

Option 1: For the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of VCS-PD submission for validation. For the second crediting period, the build margin emission factor should be updated based on the most recent information available on units already built at the time of submission of the request for renewal of the crediting period. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period.¹⁹

Capacity additions from retrofits of power plants are not included in the calculation of the build margin emission factor.

The sample group of power units *m* used to calculate the build margin is determined as per the following procedure:

- Identify the set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently ($SET_{5-units}$) and determine their annual electricity generation ($AEG_{SET-5-units}$, in MWh);
- Determine the annual electricity generation of the project electricity system, excluding power units registered as CDM project activities (AEG_{total} , in MWh). Identify the set of power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently and that comprise 20% of AEG_{total} (if 20% falls on part of the generation of a unit, the

¹⁹ The term "CDM-PDD" given in the original text was replaced by "VCS PD" for this project activity. The CDM term "DOE" has also been removed.

generation of that unit is fully included in the calculation) ($SET \geq 20\%$) and determine their annual electricity generation ($AEG_{SET \geq 20\%}$, in MWh);

- (c) From $SET_{5-units}$ and $SET_{\geq 20\%}$ select the set of power units that comprises the larger annual electricity generation (SET_{sample});

Identify the date when the power units in SET_{sample} started to supply electricity to the grid. If none of the power units in SET_{sample} started to supply electricity to the grid more than 10 years ago, then use SET_{sample} to calculate the build margin. Ignore steps (d), (e) and (f).

Otherwise:

- (d) Exclude from SET_{sample} the power units which started to supply electricity to the grid more than 10 years ago. Include in that set the power units registered as CDM project activity, starting with power units that started to supply electricity to the grid most recently, until the electricity generation of the new set comprises 20% of the annual electricity generation of the project electricity system (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) to the extent is possible. Determine for the resulting set ($SET_{sample-CDM}$) the annual electricity generation ($AEG_{SET-sample-CDM}$, in MWh); If the annual electricity generation of that set is comprises at least 20% of the annual electricity generation of the project electricity system (i.e. $AEG_{SET-sample-CDM} \geq 0.2 \times EG_{total}$), then use the sample group $SET_{sample-CDM}$ to calculate the build margin. Ignore steps (e) and (f).

Otherwise:

- (e) Include in the sample group $SET_{sample-CDM}$ the power units that started to supply electricity to the grid more than 10 years ago until the electricity generation of the new set comprises 20% of the annual electricity generation of the project electricity system (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation);
- (f) The sample group of power units m used to calculate the build margin is the resulting set ($SET_{sample-CDM->10yrs}$).

As can be seen in the excel sheet of the emission factor ("EF Colombia 2010-2012"), in this case all steps (a) to (f) need to be applied and the resulting sample group of power units m is the $SET_{sample-CDM->10yrs}$.

The build margin emissions factor is the generation-weighted average emission factor (tCO₂/MWh) of all power units m during the most recent year y for which power generation data is available, i.e. in this case the year 2011. The calculation is made as follows:

$$EF_{grid,BM,y} = \frac{\sum_m EG_{m,y} \times EF_{EL,m,y}}{\sum_m EG_{m,y}} \quad (5)$$

Where:

$EF_{grid,BM,y}$	= Build margin CO ₂ emission factor in year y (tCO ₂ /MWh)
$EG_{m,y}$	= Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	= CO ₂ emission factor of power unit m in year y (tCO ₂ /MWh)
m	= Power units included in the build margin
y	= Most recent historical year for which power generation data is available

The emission factor of each power unit m in the build margin is determined analogous as for the operating margin: For all power units m the data on fuel consumption and electricity generation is available, therefore option A1 of the tool is chosen as the most appropriate approach.

The total generation in 2012 was 59,995,370 MWh, which means that 20% of the total generation is 11,999,074 MWh. The total generation included in the BM is 12,053,308 MWh with corresponding CO₂ emissions of 1,622,468 tCO₂.

The detailed calculations are provided in the worksheet "BM" of the Excel file "EF Colombia 2010-2012".

The resulting BM emission factor is **0.2044 tCO₂/MWh**. (rounded down).

Step 6: Calculate the combined margin (CM) emissions factor

The combined margin emissions factor is calculated as follows:

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM} \quad (6)$$

Where:

$EF_{grid,OM,y}$	= Operating margin CO ₂ emission factor in year y (tCO ₂ /MWh)
$EF_{grid,BM,y}$	= Build margin CO ₂ emission factor in year y (tCO ₂ /MWh)
w_{OM}	= Weighting of operating margin emissions factor (%)
w_{BM}	= Weighting of build margin emissions factor (%)

The weighting of operating and build margin is done as indicated in the tool for the first crediting period, i.e. $w_{OM} = 0.5$ and $w_{BM} = 0.5$.

The combined margin emission factor is calculated as $EF_{grid,CM,y} = 0.3072$ tCO₂/MWh (rounded down).

The ex ante emission reductions gives

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y} = 104,500 \text{ MWh} \times 0.3072 \text{ tCO}_2/\text{MWh}$$

$$BE_y = 32,102 \text{ tCO}_2.$$

3.2 Project Emissions

As per the applied ACM0002 version 13.0.0, for hydro power project activities that result in new reservoirs, project proponents do not need to account for CH₄ and CO₂ emissions from the reservoir, if the power density of the project activity (PD) is greater than 10 W/m².

$$PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}} \quad (7)$$

Where:

PD	= Power density of the project activity (W/m ²)
Cap _{PJ}	= Installed capacity of the hydro power plant after the implementation of the project activity (W)
Cap _{BL}	= Installed capacity of the hydro power plant before the implementation of the project activity (W). For new hydro power plants, this value is zero
A _{PJ}	= Area of the reservoir measured in the surface of the water, after the implementation of the project activity, when the reservoir is full (m ²)
A _{BL}	= Area of the reservoir measured in the surface of the water, before the implementation of the project activity, when the reservoir is full (m ²). For new reservoirs, this value is zero

Since the project activity has no reservoir, this step is not required and the project emissions are zero:

$$PE_y = 0$$

3.3 Leakage

No leakage emissions are considered as provided in the methodology:

$$LE_y = 0$$

3.4 Summary of GHG Emission Reductions and Removals

Since project emissions and leakage emissions are zero, emission reductions are directly given as the baseline emissions as follows:

$$ER_y = BE_y - PE_y - LE_y = BE_y \quad (8)$$

Where:

- ER_y = Emission reductions in year y (tCO₂e/yr)
- BE_y = Baseline emissions in year y (tCO₂/yr)
- PE_y = Project emissions in year y (tCO₂e/yr)
- LE_y = Project emissions in year y (tCO₂e/yr)

Thus: $ER_y = 32,102$ tCO₂/yr

Years	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
2015	32,102	0	0	32,102
2016	32,102	0	0	32,102
2017	32,102	0	0	32,102
2018	32,102	0	0	32,102
2019	32,102	0	0	32,102
2020	32,102	0	0	32,102
2021	32,102	0	0	32,102
2022	32,102	0	0	32,102
2023	32,102	0	0	32,102
2024	32,102	0	0	32,102

Total	321,024	0	0	321,024
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4 MONITORING

4.1 Data and Parameters Available at Validation

Data Unit / Parameter:	EF_{grid,CM,y}
Data unit:	tCO ₂ /MWh
Description:	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system”
Source of data:	Calculated based on information provided by the grid operator XM.
Value applied:	0.3072 tCO ₂ /MWh.
Justification of choice of data or description of measurement methods and procedures applied:	As per the most recent “Tool to calculate the emission factor for an electricity system”
Any comment:	-

4.2 Data and Parameters Monitored

Data Unit / Parameter:	EG_{facility,y}
Data unit:	MWh/yr
Description:	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y
Source of data:	Electricity meter(s)
Description of measurement methods and procedures to be applied:	The following parameters will be measured: The meters installed will be bi-directional, taking into account both the quantity of electricity supplied by the project plant to the grid and the quantity of electricity delivered to the project plant from the grid. In this case the individual values are not required.
Frequency of monitoring/recording:	Continuous measurement and at least monthly recording. Typically, the measured data is read once every 24 hours using tele-metetering technology (remotely) and reported to XM (grid operator and administrator).

Value applied:	104,500 MWh
Monitoring equipment:	A main and a backup meter will be installed. Both will be bi-directional and have an accuracy class 0.2S.
QA/QC procedures to be applied:	Calibration tasks follow national standards and are in accordance with the calibration instructive specified in Colombian standard NTC 4,856 for electricity metering devices. There is no regulatory requirement for calibration frequency. The meters will be calibrated maximum every three years based on national standards.
Calculation method:	Not applicable
Any comment:	All data collected as part of the monitoring process is archived electronically and kept at least for two years after the end of the last crediting period.

4.3 Description of the Monitoring Plan

The Monitoring Plan basically consists in the procedures to meter the project's electricity generation. Since this is the only parameter to be measured and the emission factor is already set ex ante, the process is pretty straightforward, as described in the following.

Electricity generation of the project activity

Monitoring procedures are implemented onsite or remote, using tele-metering technology. The operational team is in charge of taking the measurements and reporting to XM. A main and backup meters are installed at the interconnection point of the project with the SIN²⁰. The energy meters in the substation are read via dedicated software every 24 hours. A daily report is then generated and sent to XM.

As indicated, there is a main electricity meter and a backup meter, which assures correct measurement in case of failures of the main electricity meters. If any other emergency would not allow measuring correctly the power generation of the plants or if data would be missing as a result of a failure in the monitoring process, no emission reductions shall be claimed during that period until guaranteeing again correct function of the meters and having reliable data.

The information is backed up by the operational team frequently. The data is included in an excel spreadsheet for emission reductions calculations on a monthly basis. All data collected as part of the monitoring process are archived electronically and kept at least for two years after the end of the last crediting period.

The following scheme shows the power plants, the substation and the metering points:

²⁰ In the Colombian regulation, this is called "commercial boundary" or in Spanish: "frontera comercial". The substation is not yet defined.

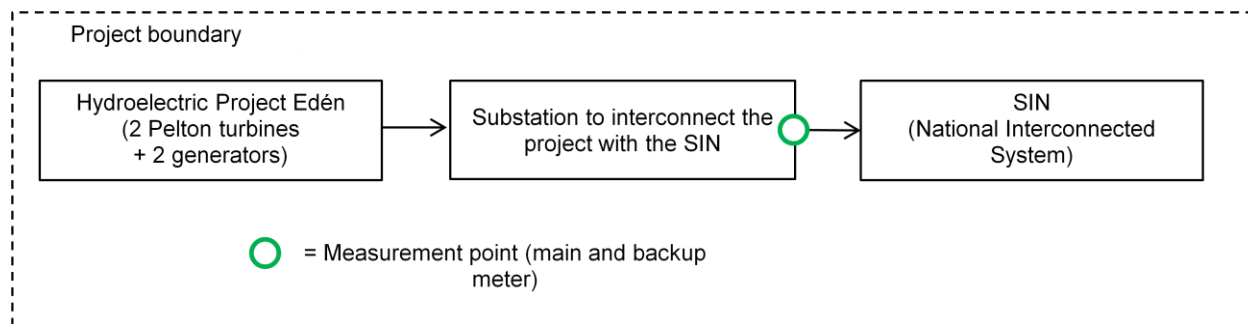


Figure 10: Simplified scheme of the monitoring.

QA/QC measures

The meters are calibrated maximum every three years. Calibration tasks follow national standards and are in accordance with the calibration instructive specified in Colombian standard NTC 4,856 for electricity metering devices. In case both meters fail, no emission reductions will be claimed during that period until having again data from the main or backup meter.

Personnel responsible for monitoring

Figure 11 shows the organizational chart of the monitoring.

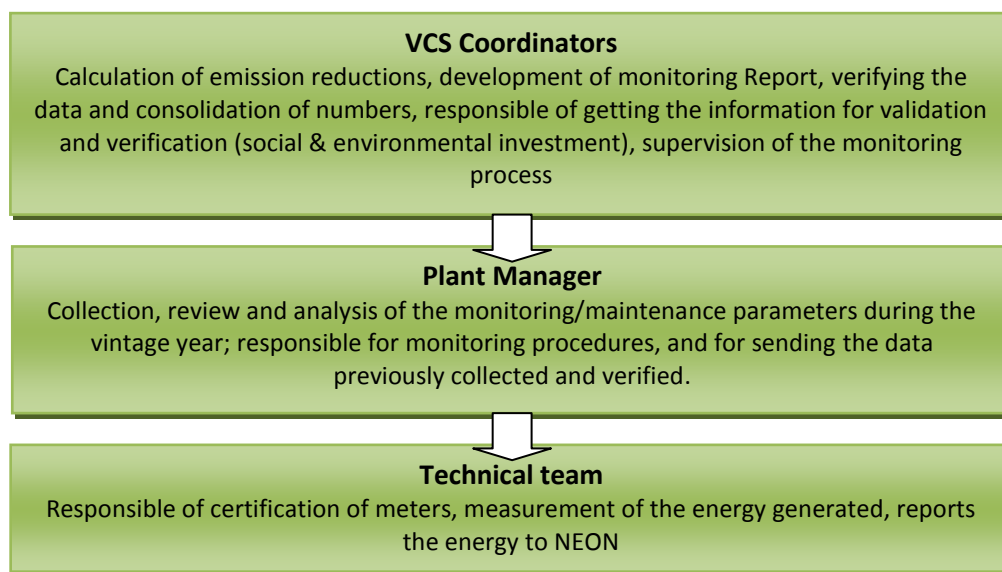


Figure 11. Operational structure of the monitoring plan

Responsible personnel:

- The CDM Coordinators supervise the monitoring process and are in charge of compiling the monitoring data in an excel spreadsheet and calculating the emission reductions of the monitoring period. They also develop the monitoring report in accordance with the VCS rules.

- The Plant Manager is responsible for verification of energy measurement. He is responsible for checking and verifying the meter readings.
- The Instrumentation and Control Engineers are responsible for electricity generation reading and for processing the energy produced by the project from the meters installed at the substation. Records of the meter are downloaded in a spreadsheet for measurement control and the data discharged from the meter is stored electronically and sent to NEON (www.xm.com.co).

Personnel who carry out monitoring tasks are familiar with the basic monitoring requirements and structures. New personnel have to participate in a basic training in order to get familiarized with the monitoring procedures. Since the main monitoring tasks, i.e. the measurement of the energy production, the calibration of energy meters, and the reporting of the energy generation, are carried out independently from the VCS as part of the daily operation, no specific training is required. The VCS Coordinators are supported by an external consultant if necessary in order to assure correct application of the monitoring procedures. They also carry out corrective actions if any inconsistency is identified and train the Plant Manager and the Technical team, if necessary.

5 ENVIRONMENTAL IMPACT

The law 99 from 1993 provides the general requirements for issuing environmental licenses and permits and defines the role of the Environmental Ministry and the Regional Autonomous Corporations for the licensing process.

Article 49 of the law 99 of 1993 and article 3 of the decree 1220 of 2005 indicates that the execution of works, establishment of industries or the development of any activity that in accordance with the law or any applicable regulation could produce significant impact on natural resources or the environment or introduce considerable modifications to the landscape, require an environmental license.

As per Article 52 of law 99 and article 8 of decree 1220, projects with less than 100 MW do not fall under the responsibility of the Environmental Ministry but the Regional Autonomous Corporation – in this case CORPOCALDAS.

An Environmental Impact Assessment (EIA) was carried out in order to obtain the environmental licence. The general objective of the study is to evaluate the environmental, physical, biotical and socio-economical components in the area of influence, and the necessary natural resources for the construction and operation.

Environmental Impact Assessment (EIA)

The EIA serves as an instrument for decision making regarding the environmental impacts generated by the project corresponding to the measures in the environmental management and monitoring plan to prevent, correct, compensate and mitigate the impacts. The specific objectives are:

- Identify and locate the area of influence (regional, local and direct) of the project
- Characterize the environment of the direct and indirect area of influence
- Evaluate the positive and negative environmental impacts during the construction and operational phases
- Elaborate the Environmental Management and Monitoring Plan\
- Construct the Contingency Plan to evaluate the project risks
- Establish the actions after the project's lifetime
- Propose a techno-economical plan for the investment of the 1%

In general, the conclusions of the EIA were positive and allowed the issuance of the environmental license on May 4, 2011 (see Resolution 173 “Licencia Ambiental El Eden.pdf”). A small modification of the project design was made subsequently, upon which the modified EIA was presented and a second environmental license emitted on August 5, 2013 (see Resolution 279 “Modification Licencia Ambiental El Eden.pdf”).

6 STAKEHOLDER COMMENTS

The project has started the local stakeholder consultation process in August 2010 and since then carried out a series of activities to inform and take into account the communities in the area of direct and indirect influence. A first stakeholder round was organized from August 2010 to May 2011 in the different communities. A second round was carried out from June to August 2011. And finally a third round with different meetings and visits during the period from March to May 2012 was carried out. The Figure 12 shows the participants in each round.

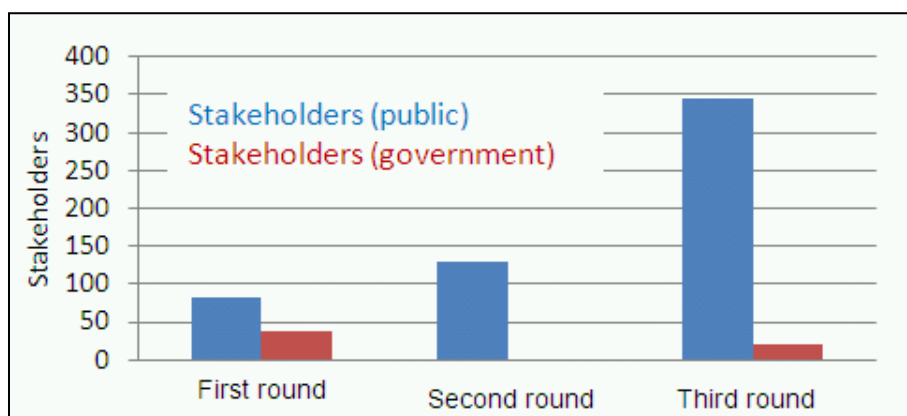


Figure 12: Stakeholder participation in the initial rounds.

During the third stakeholder round, a total over 350 persons of the public and governmental entities participated in different meetings and workshops.

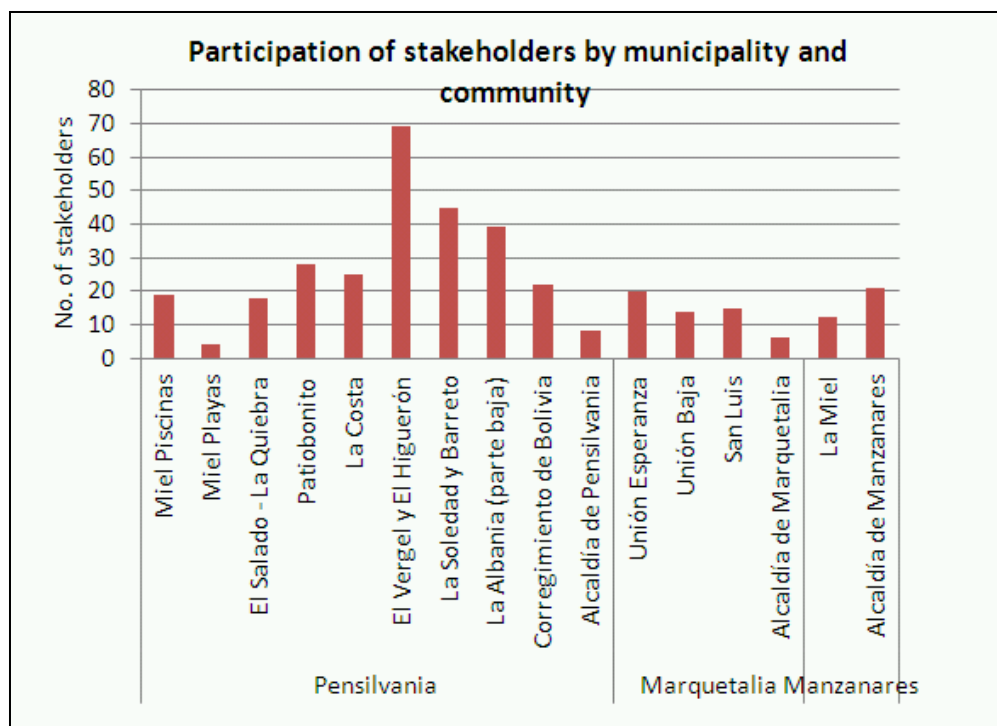


Figure 13: Stakeholder participation in the initial rounds.

In the following a major event of the stakeholder process including the environmental impacts assessment, environmental management and modification of the environmental license is described. The meeting was held on 14th of April 2013 at the John F. Kennedy School of the community of Bolivia which is close to the project location. The stakeholders were invited by letters of invitation to the presidents and members of the boards of communitarian actions that further informed the communities.

The meeting included the following main topics:

1. Opening and presentation
2. Objectives of the meeting
3. Participants of the project
4. Stages of the project
5. History and modification of the environmental license
6. Project description
7. Aspects and activities that generate impacts
8. Environmental impacts and impact management
9. Assessment and participation of stakeholders

Figure 14 shows some impressions of the meeting.



Figure 14: Impressions of the local stakeholder meeting.

The following is a summary of the questions and answers that were made during the meeting (only the most important topics included in this list):

Q: Extend to which the non qualified workers required for the construction will be from the region.

A: All non qualified workers used for the construction will be locally hired.

Q: Improvement of the roads in the area.

A: The management is assessing to make maintenance of other roads in the close area to the project.

Q: Availability of water after implementation of the project and requirement to sign contracts for water security.

A: The environmental license is the security which is approved by COROCALDAS based on the detailed studies. Besides, it is clarified and explained that there is an environmental management plan that defines the required actions to assure minimization of the impacts.

The final commitment and agreement of the meeting was to continue the stakeholder process and to review the requests of the communities.